

SOMRAJ GAUTAM

Researcher in Document AI & Vision / Large-Language Models

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RESEARCH INTERESTS

Document Image Analysis • Table Understanding & Interpretation • Visual Question Answering • Active Learning • Multilingual Document AI • LLM / VLM • ASEAN Language Processing • NLP / NLU

EDUCATION

- Feb 2021 - Present **PhD in Computer Science and Engineering - IIT Jodhpur, India**
Advisor: Prof. Gaurav Harit
Thesis: Learning to See, Read, and Reason over Tables: A Unified Framework for Document Intelligence
Research Committee: Dr. V. Narayanan, Dr. Rajendra Nagar, Dr. Yashaswi Verma
- Aug 2018 - May 2020 **M.Tech in Computer Science and Engineering - IIT Dhanbad, India**
Advisor: Dr. Tanusree Kaibartta
Thesis: Test Cost Optimization of Core-Based Non-Stacked ICs
- Aug 2013 - May 2017 **B.Tech in Computer Science and Engineering - BIET Jhansi, India**
Advisor: Prof. A. K. Solanki

PUBLICATIONS

Peer-Reviewed Conference & Workshop Papers

- 1) **Somraj Gautam**, A. Bhandari, G. Harit. [TabComp: A Dataset for Visual Table Reading Comprehension](#). **NAACL 2025 (Core-A)**
OCR-free table QA benchmark with structured reasoning.
- 2) **Somraj Gautam**, N. Purohit, G. Harit. [Table Detection with Active Learning](#). **ICDAR 2025 (Core-A)**
Hybrid uncertainty + diversity sampling-based active learning for table detection.
- 3) **Somraj Gautam***, A. Subramanyam*, A. Bhandari, G. Harit. [Mind the \(Language\) Gap: Probing Numerical and Cross-Lingual Limits of LVLMs](#). **EMNLP 2025 Workshop (Core-A*)**
MMCRICBENCH-3K: cross-lingual table VQA benchmark (Hindi/English).
- 4) **Somraj Gautam**, A. Dravichi, G. Harit. [INDOTABVQA: Cross-Lingual Table VQA on Real-World Bahasa Indonesian Document Images](#). **ACL 2026 (Core-A*)**
First real-world Indonesian cross-lingual table VQA dataset showing significant gains via fine-tuning and spatial priors.
- 5) **Somraj Gautam**, A. Bhandari, G. Harit. [Fragmented Financials: FinTab-Multi and the Challenge of Relational Reasoning Across Multiple Document Tables](#) **Under Submission EMNLP 2026 (Core-A*)**
Benchmark + Method for robust reasoning over multiple financial table images.

- 6) **Somraj Gautam***, A. Dravichi*, G. Harit. *IndoPara: When Similarity is Not Enough for Paraphrase Identification* **Under Preparation for TMLR Journal**
- 7) **Somraj Gautam**, S.Khanna. *Can Vision-Language Models Trust What They Read? A Benchmark for Financial Table Integrity Verification* **Under Submission ACCV 2026**

Journal Papers (Under Review)

- 1) **Somraj Gautam**, G. Harit. *Cycle-of-Thought: Self-Evaluative Iterative Reasoning for Table Visual Question Answering in Document Images*. **New Generation Computing (Under Review, IF: 2.3)**
Developed Self-RefineVQA, a training-free inference-time self-correction framework enabling VLMs to generate, critique, and iteratively refine reasoning for table-based VQA. Achieves consistent accuracy gains for small-scale models without supervised fine-tuning.

TEACHING & MENTORING EXPERIENCE

Feb 2021 - Present	Teaching Assistant - IIT Jodhpur
Aug 2021 - Present	Teaching Assistant - IIT(ISM) Dhanbad
May 2024 - May 2025	Mentored 1 MTech Student, which resulted in published work in ICDAR 2025
May 2025 - May 2026	Mentoring BTech Student, submitted to ACL 2026, and preparing for EMNL 2026.

RESEARCH & WORK EXPERIENCE

Feb 2021 - Present	Junior/Senior Research Fellow (JRF/SRF) - IIT Jodhpur Conducting PhD research on table interpretation in document images. Developing novel frameworks for table detection, table QA, active learning, and multilingual document understanding. Proposed Self-RefineVQA and contributed to multiple top-tier publications.
Jan 2021 - Sep 2021	Project Developer - National Informatics Center (NIC), Dantewada Developed Sanket, a Sign Language to Speech Converter using InceptionV3 to improve accessibility for the hearing-impaired community.

AWARDS & HONOURS

- Best Research Presentation - TLS Workshop 2025, IIT Delhi.
- Travel Grant - Summer School on Generative AI for Text, IIT Gandhinagar, 2024.
- Travel Grant - ACM India Winter School on Optimization for ML & Operations Research, 2022.
- Travel Grant - 6th Summer School on AI (Computer Vision & ML), IIIT Hyderabad.
- Ministry of Education (MoE) Fellowship Awardee - IIT Jodhpur, 2021.
- Ministry of Education (MoE) Fellowship Awardee - IIT(ISM) Dhanbad, 2018.
- Prime Minister's Scholarship Scheme (PMSS), Ministry of Defense, Government of India, 2013.
- International Mathematics Olympiad - Rank 4, 2010.

ACADEMIC SERVICE

Reviewer:

- ACL 2026
- NeurIPS 2026
- Journal: SN Computer Science (Springer).

Presentations

- S. Gautam et al.: An Approach of Bin Packing Technique for Test Cost Optimization of Core-Based Non-Stacked ICs. Workshop on RTL and High-Level Testing (WRTLTL-19), Asian Test Symposium, Kolkata, India, 2019. (Oral Presentation).

TECHNICAL SKILLS

Languages	Python, C++
ML Frameworks	PyTorch, TensorFlow, scikit-learn
Research Tools	HuggingFace Transformers, Weights & Biases, OpenCV
Infrastructure	Google Colab, Kaggle, Git, Overleaf

NATIONAL EXAMINATIONS & QUALIFICATIONS

- GATE 2020 (Computer Science)
- GATE 2018 (Computer Science)
- 3rd Rank - National Scholarship Test (NST), Made-Easy, 2017
- JEE Advanced 2013 | JEE Mains 2013

REFERENCES

1. Prof. Gaurav Harit
Professor
Department of Computer Science and Engineering
IIT Jodhpur
2. Prof. V. Narayanan
Professor
Department of Physics
IIT Jodhpur
3. Dr. Rajendra Nagar
Assistant Professor
Department of Electrical Engineering
IIT Jodhpur
4. Dr. Yashaswi Verma
Associate Professor
Department of Computer Science and Engineering
IIT Jodhpur
5. Dr. Tanushree Kaibartta
Assistant Professor
Department of Computer Science and Engineering
IIT (ISM) Dhanbad